MBAG – 407 – Machine Learning for Managers

Project

End-to-End Machine Learning Pipeline

Spotify Song Popularity Prediction

Dataset: Spotify Tracks Dataset (Kaggle)

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About the Dataset:

Name: 🎹 Spotify Tracks Dataset

Source: Kaggle

Link: https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset/code

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Executive Summary

The music industry spends millions of dollars signing artists, producing albums, and funding marketing campaigns — often with no data-driven basis for deciding which songs deserve that investment. Most songs released on Spotify never find an audience. This project answers a direct question that every record label faces: can we predict whether a song will become a chart-topper, a mainstream track, or remain undiscovered — before it is released?

Using Spotify's internal audio measurements for 114,000 songs across 114 genres, a machine learning model was built and tested to classify each song into one of three popularity categories: High, Medium, or Low. The best-performing model — a Random Forest classifier — achieved 74% accuracy and a Weighted F1-Score of 72.7% on songs it had never seen before. This means the model correctly identifies nearly three out of every four songs' commercial potential, and does so in under 10 seconds.

The findings from this project provide a practical, data-backed screening tool that can guide investment decisions before a single rupee of marketing budget is committed.

The Business Problem

Spotify assigns every song a popularity score from 0 to 100 based on total streams and recency of listening. An analysis of 114,000 songs reveals a stark reality: only 7.5% of all songs on the platform qualify as genuinely popular (score above 65). Nearly 60% of songs score below 40 — they are essentially undiscovered. The remaining 33.5% sit in a mainstream middle ground.

For record labels and music producers, this means that for every chart-topping hit, approximately 13 songs fail to gain meaningful traction. The cost of backing the wrong song is real — wasted production budgets, marketing spend, and opportunity cost. The cost of missing a genuine hit is equally real — a competitor signs the next chart-topper while the label passes.

This project builds a model that screens songs on audio characteristics alone, giving decision-makers a data-backed second opinion before investment decisions are made.

What the Data Revealed

Before building any model, the data was analysed to understand what actually separates chart-toppers from undiscovered tracks. Seven findings stand out:

- **Song duration is the strongest predictor of commercial success.** The length of a song — measured in minutes — was the single most important feature the model identified. Songs that fall within the streaming platform's "sweet spot" of approximately 3 to 4 minutes consistently outperform shorter or significantly longer tracks. This is an actionable production decision that can be made before a song is released.
- **Emotional quality and production loudness follow closely.** Songs with higher valence — meaning a more positive, upbeat emotional character — and those mastered to competitive loudness levels (approximately -4 to -6 dBFS) appear disproportionately among highly popular tracks. These are also decisions made in the studio, before release.
- **Genre positioning matters, but less than audio quality.** Songs in the Pop & Mainstream category achieve a High popularity rate of 12.1% — roughly 1.6 times the dataset average of 7.5%. Rock & Metal follows at 9.6%, while Electronic & Dance, Hip-Hop, R&B & Urban, and Other genres all sit below the average. Genre is a meaningful signal, but the audio characteristics of a song outweigh genre positioning in predicting commercial outcomes.
- **Explicit content correlates with higher chart performance.** Explicit songs achieve a high popularity at 13% compared to 7% for non-explicit tracks. This does not mean explicit content causes popularity — it reflects that the genres and production styles that dominate chart performance tend to carry explicit content. Labels that restrict explicit content across the board may inadvertently reduce their commercial reach in the most competitive genres.
- **Instrumental tracks rarely chart.** Songs with high instrumental scores — meaning little to no vocal content — cluster overwhelmingly in the Low popularity category. Instrumental releases require an entirely different distribution strategy: sync licensing for film and television, specialised editorial playlists, and background music platforms rather than mainstream streaming campaigns.
- **Energy and acoustic style are mutually exclusive at scale.** Across 114,000 songs, high-energy tracks are rarely acoustic, and acoustic tracks are rarely high-energy. This confirms that the electronic vs. acoustic character of a song is a meaningful dimension of commercial positioning, not just a stylistic choice.
- **Popularity cannot be predicted with simple rules.** The gap between the simplest model tested (Logistic Regression, 45.9% F1-Score) and the best model (Random Forest, 72.4% F1-Score) is over 26 percentage points. No single audio feature reliably predicts commercial success on its own — it is the combination of duration, emotional quality, vocal presence, loudness, tempo, and genre positioning working together that determines outcomes. This is why human intuition alone, which tends to focus on one or two features, consistently underperforms data-driven approaches.

Model Performance

Four different machine learning models were trained and compared on a validation set of 17,103 songs. The primary metric used for comparison was Weighted F1-Score rather than raw accuracy, because accuracy alone is misleading when 59% of all songs are of low popularity. A model that simply predicted "Low" for every song would appear to be 59% accurate while being completely useless for the business.

Model	Accuracy	Weighted F1-Score
Logistic Regression	42.0%	45.9%
LinearSVC	59.6%	50.9%
XGBoost	65.9%	62.1%
Random Forest	**73.7%**	**72.4%**

The Random Forest model was selected as the best performer and evaluated on a separate held-out test set of 17,100 songs that was never used during training or model selection. On this final evaluation, the model achieved 74% accuracy and a Weighted F1-Score of 72.7% — confirming the result is genuine and not a product of overfitting to the training data. A stability check across five independent data splits (k-Fold Cross Validation) showed a mean score of 71.1% with a standard deviation of only 0.5%, confirming the model performs consistently regardless of which songs it is trained on.

Breaking down performance by popularity class on the test set:

- **Low popularity songs: 89.4% correctly identified** — the model is highly reliable at flagging tracks unlikely to gain traction.
- **Medium popularity songs: 54.2% correctly identified** — harder to separate from Low, as expected given the continuous nature of the underlying popularity score.
- **High popularity songs: 40.5% correctly identified** — meaning the model correctly flagged approximately 518 out of 1,279 actual chart-toppers. Given that only 7.5% of songs are in this category, this represents significant commercial value in a real screening workflow.

The dramatic failure of Logistic Regression (45.9% Weighted F1) is itself an important finding: it confirms that the relationship between audio features and popularity is fundamentally non-linear. No simple rule of the form "if danceability is high and energy is high, the song will chart" can capture what the data shows. Only models capable of learning complex, combined patterns — like Random Forest — can perform at a commercially useful level.

Business Recommendations

1. **Adopt the model as a pre-release screening tool.** Before committing to a full marketing campaign, run every new track through the model. Songs predicted as Medium or High with reasonable confidence should receive playlist investment and

promotion. Songs predicted as Low should be released but routed toward organic discovery rather than paid campaigns — saving budget for stronger candidates.

2. **Make duration a first-order production decision.** Since song length is the strongest single predictor the model identified, producers should explicitly target the 3–4 minute range during the arrangement and editing phase. This is one of the few commercially significant audio attributes that can be directly adjusted without changing the artistic character of the song.
3. **Invest in mastering quality and emotional positivity.** Valence (upbeat, positive feel) and loudness (competitive mastering) are the next strongest predictors after duration. Producers should not cut corners on mastering, and A&R teams should be aware that emotionally positive songs have a measurable commercial advantage in this dataset.
4. **Route instrumental releases differently.** Any track flagged as instrumental should not receive a mainstream streaming campaign. Instead, route these releases toward sync licensing opportunities (film, television, advertising), lo-fi and focus music playlists, and background music platforms where instrumental content is specifically sought.
5. **Do not apply blanket restrictions on explicit content.** For Pop, Hip-Hop, and electronic artists, the data shows explicit songs are significantly more likely to chart. Restricting explicit content as a default policy should be a deliberate strategic trade-off, not a blanket rule.

Limitations

This model is a tool to support decisions, not replace them. Three limitations are important to understand before deploying it:

1. **The model cannot account for artist fame or label power.** A globally recognised artist's average song will outperform an unknown artist's excellent song every time. Audio features measured at recording time do not capture this. Adding artist-level data — follower counts, historical chart performance, label affiliation — would significantly improve predictions.
2. **Popularity scores change over time.** A song that scored 80 in 2022 may score 45 today as newer songs accumulate streams. The model was trained on a historical snapshot and should be retrained every six to twelve months using updated popularity data to remain relevant.
3. **Viral and social media effects are invisible to the model.** A song can be audio-average and become a global hit because of a TikTok trend or a placement in a major television series. These events cannot be predicted from audio features alone. The model is best understood as a baseline screen — it identifies songs with the right audio profile for mainstream success, not songs that will necessarily go viral.